# A Modified Newton Method for Radial Distribution System Power Flow Analysis

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**Abstract** A modified Newton method for radial distribution systems is derived in which the Jacobian matrix is in $UDU^T$ form, where $U$ is a constant upper triangular matrix depending solely on system topology and $D$ is a block diagonal matrix. With this formulation, the conventional steps of forming the Jacobian matrix, LU factorization and forward/back substitution are replaced by back/forward sweeps on radial feeders with equivalent impedances. Tests on several large distribution systems ranged from 490 to 1651 in nodes, 0.15 to 5.48 in r/x ratio and 0.0004 Ω to 3.07 Ω in line impedance have shown that the proposed method is as robust and efficient as the back/forward sweep method[8]. The proposed method can be applied to other applications, such as state estimation. The proposed method can also be extended to the solution of systems with loops, dispersed generators and three phase (unbalanced) representation.

**Key Words:** Power Flow, Radial Distribution Systems, State Estimation, Dispersed Generators, Looped Distribution Systems.

## I. Introduction

Power flow is the most fundamental numerical algorithm for power system analysis. In 1967, Tinney and Hart developed the classical Newton based power flow solution method[1]. Later work by Stott and Alsac[4] made the fast decoupled Newton method as well as its various alternatives a kind of standard power flow method for EMS systems. Many other applications have been developed based on the Newton formulation since then.

Even though the fast decoupled Newton method works well for transmission systems, its convergence performance is poor for most distribution systems due to their high r/x ratio which deteriorates the diagonal dominance of the Jacobian matrix. For this reason, several non-Newton type of methods have been proposed[5-8]. Their algorithms all consist of back/forward sweeps on a ladder system. The formulation and the algorithm of these methods are different from the Newton's power flow method, which makes these methods hard to be extended to other applications, such as the state estimation and the optimal power flow, in which the Newton method seems more appropriate. In fact, the distribution state estimation methods proposed in [13-16] are all based on the Newton method.

Recently, a fast decoupled power flow method has been proposed in [9]. This method orders the "laterals" instead of "buses" into layers, thus reduces the problem size to the number of laterals, then assumes initial end voltage for all laterals. The iteration starts from the first lateral by using the method proposed in [6]. The voltage mismatch obtained from this lateral is applied to correct not only the end voltage of this lateral but also the end voltage of laterals of the next layer. The algorithm converges when all voltage mismatches are within a certain tolerance. Using lateral variables instead of node variables makes this method efficient for a given system topology, but it may add some overhead if the system topology is changed regularly, which is common in distribution systems due to switching operations.

The purpose of this paper is to derive a modified Newton method for radial distribution systems without reducing the problem size, yet still capable of achieving robust convergence and high efficiency. Specifically, this paper aims at deriving a Newton formulation in which the Jacobian matrix is in $UDU^T$ form, where $U$ is a constant upper triangular matrix depending solely on system topology and $D$ is a block diagonal matrix resulting from the radial structure and special properties of the distribution system. With this formulation, the conventional Newton algorithm of forming the Jacobian matrix, LU factorization and forward back substitution can be replaced by back /forward sweeps on radial feeders with equivalent impedances.

To assist in presenting the essential difference between the proposed method and other methods, the following definitions are used:

(1) Conventional Newton method: a method in which the derivative of the power flow equation, i.e., the Jacobian matrix (exact or approximate), is used to

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determine the search direction, and forward/back substitutions on the LU factors of the Jacobian matrix are used to calculate the incremental correction of the state variables.

(2) Modified Newton method: the proposed method in which an approximate Jacobian matrix in $UDU^T$ form is used to determine the search direction, and the linearized power flow equation based on this Jacobian matrix is used as basis for back/forward sweeps on a radial network to calculate the incremental correction of the state variables.

(3) Back/forward sweep method: a method in which the derivative of the power flow equation is not used, instead basic circuit laws, i.e., Ohm's Law, KVL and KCL (or the generalized KCL for power summation) are used as basis for back/forward sweeps on a radial network to calculate the incremental correction of the state variables.

The advantage of the proposed method includes: (1) it is a Newton method thus it can be extended to other applications such as the state estimation, (2) the Jacobian matrix in $UDU^T$ form does not need to be explicitly formed, and the back/forward sweeps are directly based on the linearized power flow equations. Thus possible ill-conditions associated with the Jacobian matrix and its LU factors [2][3] are completely avoided, and (3) test results have shown that it is as robust and efficient as the back/forward sweep method[8].

This paper is organized as follows: In section II, the basic circuit theory is applied to radial distribution systems and to explain the back/forward sweep algorithm. The proposed Newton method is derived in section III, and its robustness and computational efficiency are compared with the back/forward sweep method in section IV. In section V, the proposed Newton method is applied to the state estimation problem of distribution systems, and is extended to the solution of systems with loops, dispersed generators and three phase (unbalanced) representation. At the end, Appendix A provides a brief review of the basic circuit theory in matrix form.

## II. Basic Circuit Theory in Distribution Systems

For a linear, time-invariant RLC circuit with a sinusoidal voltage source, the basic circuit theory can be expressed as (see Appendix A for details):

Ohm's Law: $\dot{I}_b = Y_b \dot{V}_b$ (1)

KCL: $A\dot{I}_b = \dot{I}_n$ (2)

KVL: $B\dot{V}_b = 0$ (3)

For fundamental frequency power flow calculations, a distribution system is always modeled as a linear, time-invariant RLC circuit. Earth is always treated as a reference node. For a radial distribution system with n nodes and without shunt branches, the number of branches is n-1. Therefore, the dimension of matrix $A$ is n x (n-1).

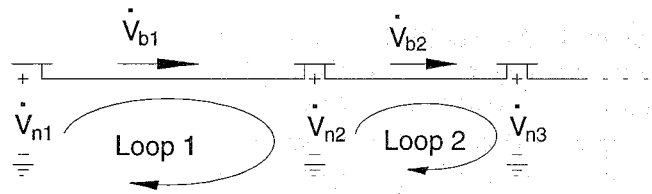


**Figure 1**

The independent loop for radial distribution systems can always be formed by a branch with its two shunt branches. Since shunt branches are usually neglected in modeling distribution lines, a fictitious shunt branch can be placed with branch voltage to be the nodal voltage as shown in Figure 1, and (3) can be written as:

$$\dot{V}_b = A^T \dot{V}_n \tag{4}$$

Combining (1), (2), and (4), we have:

$$AY_b A^T \dot{V}_n = \dot{I}_n \tag{5}$$

By knowing the nodal voltage at one node, assuming it is the first node for convenience, and nodal current injections at the other n-1 nodes, (6) can be derived from (5) for solving the remaining n-1 unknown nodal voltages:

$$A_{n-1} Y_b \begin{pmatrix} A_s^T & A_{n-1}^T \end{pmatrix} \begin{pmatrix} \dot{V}_s \\ \dot{V}_{n-1} \end{pmatrix} = \dot{I}_{n-1} \tag{6}$$

where $A = \begin{bmatrix} A_s \\ A_{n-1} \end{bmatrix}$, $\dot{V}_n = \begin{bmatrix} \dot{V}_s \\ \dot{V}_{n-1} \end{bmatrix}$, and $\dot{I}_n = \begin{bmatrix} \dot{I}_s \\ \dot{I}_{n-1} \end{bmatrix}$. Note matrix $A_{n-1}$ is a square matrix.

Since every branch is always directed away from one node and towards the other node, we have:

$$A^T e_n = 0 \tag{7a}$$

or

$$A_s^T + A_{n-1}^T e_{n-1} = 0 \tag{7b}$$

where $e_n$ and $e_{n-1}$ are unity column vectors with dimension n and n-1 respectively. Then (6) can be simplified as:

$$A_{n-1} Y_b A_{n-1}^T (\dot{V}_{n-1} - \dot{V}_s e_{n-1}) = \dot{I}_{n-1} \tag{8}$$

$A_{n-1} Y_b A_{n-1}^T$ is the Nodal Admittance Matrix which most power engineers are familiar with. In other words, for a radial system without shunt branches, the Nodal

Admittance Matrix is formed as the product of three **square** matrices. It is interesting to note that the back /forward sweep method proposed in [5] for a radial system is exactly the same as solving $\dot{V}_{n-1}$ from (8) if we organize (8) as follows:

$$A_{n-1}\dot{I}_L = \dot{I}_{n-1} \tag{8a}$$

$$Y_b A_{n-1}^T(\dot{V}_{n-1} - \dot{V}_s e_{n-1}) = \dot{I}_L \tag{8b}$$

Solving $\dot{I}_L$ from (8a) is equivalent to the 'Backward Sweep', and solving $\dot{V}_{n-1}$ from (8b) is equivalent to the 'Forward Sweep'. This observation is very important as it motivated us to derive a Jacobian matrix in $UDU^T$ form and a back/forward sweep algorithm for the Newton method.

## III. The Modified Newton Method

In this section we will show that under the following assumptions:

small voltage difference between two adjacent nodes,
no shunt branches,

the Jacobian matrix for a radial system is formed as $UDU^T$, where $U$ is a constant upper triangular matrix depending solely on system topology and $D$ is a block diagonal matrix. The first assumption above is valid, since typical distribution lines are short and power flows are not high. The second assumption is not valid if there exist shunt capacitor banks, constant impedance loads, as well as non-negligible shunt admittance of distribution line models($\pi$ model). However, all the shunt branches can be converted to node power injections using initial and updated node voltages. Similar conversions have been used in all back/forward sweep methods[5-8], and the fast decoupled method[9].

In the conventional Newton method[1], the power flow problem is to solve (9) for $\Delta\theta$ and $\Delta V$:

$$\begin{bmatrix} H & N \\ J & L \end{bmatrix}\begin{bmatrix} \Delta\theta \\ \Delta V / V \end{bmatrix} = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} \tag{9}$$

where

$$H_{ij} = -V_i V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) \quad j \neq i \tag{10}$$

$$H_{ii} = V_i \sum_{j\in i, j\neq i} V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) \tag{11}$$

$$N_{ij} = -V_i V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \quad j \neq i \tag{12}$$

$$N_{ii} = -V_i \sum_{j\in i, j\neq i} V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) - 2V_i^2 G_{ii} \tag{13}$$

$$J_{ij} = V_i V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \quad j \neq i \tag{14}$$

$$J_{ii} = -V_i \sum_{j\in i, j\neq i} V_j (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \tag{15}$$

$$L_{ij} = -V_i V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) \quad j \neq i \tag{16}$$

$$L_{ii} = -V_i \sum_{j\in i, j\neq i} V_j (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) + 2V_i^2 B_{ii} \tag{17}$$

$G_{ij} + jB_{ij}$ is the entry of nodal admittance matrix.

Since the voltage difference between two adjacent nodes is small as well as $G_{ii} + jB_{ii} = -\sum_{j\in i, j\neq i}(G_{ij} + jB_{ij})$, for systems without shunt branches, the Jacobian matrix can be approximated as:

$$H_{ij} \approx V_i V_j B_{ij}\cos\theta_{ij} \quad j \neq i \tag{18}$$

$$H_{ii} \approx -V_i \sum_{j\in i, j\neq i} V_j B_{ij}\cos\theta_{ij} \tag{19}$$

$$N_{ij} \approx -V_i V_j G_{ij}\cos\theta_{ij} \quad j \neq i \tag{20}$$

$$N_{ii} \approx V_i \sum_{j\in i, j\neq i} V_j G_{ij}\cos\theta_{ij} \tag{21}$$

$$J_{ij} \approx V_i V_j G_{ij}\cos\theta_{ij} \quad j \neq i \tag{22}$$

$$J_{ii} \approx -V_i \sum_{j\in i, j\neq i} V_j G_{ij}\cos\theta_{ij} \tag{23}$$

$$L_{ij} \approx V_i V_j B_{ij}\cos\theta_{ij} \quad j \neq i \tag{24}$$

$$L_{ii} \approx -V_i \sum_{j\in i, j\neq i} V_j B_{ij}\cos\theta_{ij} \tag{25}$$

Equations (18) to (25) show that matrices $H$, $N$, $J$, and $L$ all have the same properties (symmetry, sparsity pattern) as the Nodal Admittance Matrix, hence they can be formed as:

$$H = L = A_{n-1} D_B A_{n-1}^T \tag{26}$$

$$J = -N = A_{n-1} D_G A_{n-1}^T \tag{27}$$

where $D_B$ and $D_G$ are diagonal matrices with diagonal entries to be $V_i V_j B_{ij}\cos\theta_{ij}$ and $V_i V_j G_{ij}\cos\theta_{ij}$ respectively. Therefore (9) can be rewritten as:

$$\begin{bmatrix} A_{n-1} & \\ & A_{n-1} \end{bmatrix}\begin{bmatrix} D_B & -D_G \\ D_G & D_B \end{bmatrix}\begin{bmatrix} A_{n-1}^T & \\ & A_{n-1}^T \end{bmatrix}\begin{bmatrix} \Delta\theta \\ \Delta V / V \end{bmatrix} = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} \tag{28}$$

It is also noted that if nodes and branches are ordered appropriately, $A_{n-1}$ is an upper triangular matrix with all diagonal entries to be 1 and all non-zero off-diagonal entries to be -1. One way to achieve such an $A_{n-1}$ is ordering branches by layers away from the root node (source bus)[5]. This ordering scheme is adopted here. The direction of each branch is towards the root node. The node ordering is proceeded simultaneously with the branch ordering. The branch 'from side' node number is the same as the branch number, as illustrated in Figure 2. The node to branch incidence matrix of Figure 2 is given in (29).

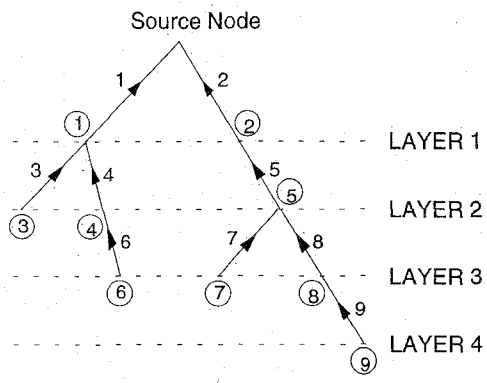


**Figure 2. A Simple Radial Distribution System**

$$A_{n-1} = \begin{bmatrix} 1 & & -1 & -1 & & & & & \\ & 1 & & & -1 & & & & \\ & & 1 & & & & & & \\ & & & 1 & & -1 & & & \\ & & & & 1 & & -1 & -1 & \\ & & & & & 1 & & & \\ & & & & & & 1 & & \\ & & & & & & & 1 & -1 \\ & & & & & & & & 1 \end{bmatrix} \tag{29}$$

By now we have shown that the Jacobian matrix can be formed as the product of three square matrices (28), same as the Nodal Admittance Matrix in (8). Next, we will show that (28) can be solved by back/forward sweeps as well. Let's define:

$$\dot{E} = \Delta\theta + j\Delta V / V \tag{30}$$

$$\dot{S} = \Delta P + j\Delta Q \tag{31}$$

$$\dot{W} = D_B + jD_G \tag{32}$$

then (28) can be written as

$$A_{n-1}\dot{W}A_{n-1}^T\dot{E} = \dot{S} \tag{33}$$

or

$$A_{n-1}\dot{S}_L = \dot{S} \tag{34}$$

$$\dot{W}A_{n-1}^T\dot{E} = \dot{S}_L \tag{35}$$

where (34) is the back sweep and (35) is the forward sweep.

To solve $\dot{E}$ in (35) in the forward sweep, the diagonal matrix $\dot{W}$ can be inverted for each line. The diagonal in $\dot{W}^{-1}$ is denoted as the equivalent line impedance:

$$Z_{eq} = R_{eq} + jX_{eq} \tag{36}$$

where

$$R_{eq} = X_{ij} / (V_i V_j \cos\theta_{ij}) \tag{37}$$

$$X_{eq} = R_{ij} / (V_i V_j \cos\theta_{ij}) \tag{38}$$

$R_{ij}$ and $X_{ij}$ are resistance and reactance of line i-j respectively. An algorithm based on (34) and (35) has been developed. Figure 3 is the flow chart of the proposed algorithm. Note that matrix $A_{n-1}$ has never been formed in the program since its non-zero entries are either -1 or 1.

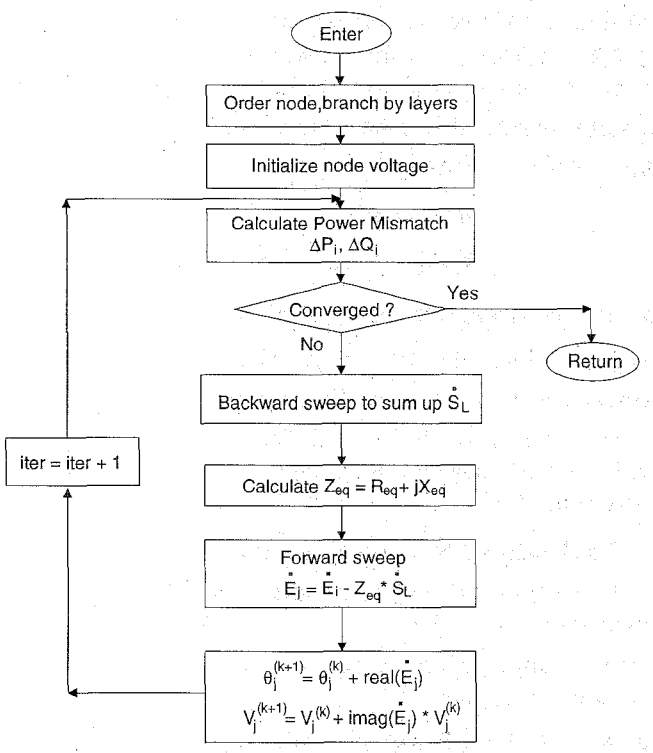


**Figure 3. Flow Chart of the Proposed Algorithm**

## IV. Test Results

The proposed algorithm shown in Figure 3 has been tested using typical primary feeders of various sizes in the PG&E distribution systems. Both capacitors and regulators are included in the test systems, but the automatic controls for switching capacitors on/off, and the automatic tap adjustment function for regulators are not modeled in the test. Summer peak loading conditions are used for all test systems. All the loads are modeled as lumped constant power load. Table 1 lists the attributes of these test systems. It is seen that all three test systems have wide ranges of r/x ratio and line impedance.

**Table 1 Attributes of Test Systems**

| Test System | No. of Nodes | Vol. Level | r/x ratio | | \|Zij\| (ohms) | |
|---|---|---|---|---|---|---|
| | | | Max | Min | Max | Min |
| 1 | 490 | 12kV | 5.06 | 0.15 | 3.07 | 0.0012 |
| 2 | 722 | 12kV | 5.06 | 0.26 | 2.43 | 0.0004 |
| 3 | 1651 | 21kV | 5.48 | 0.15 | 2.76 | 0.0005 |

The major purpose of the test is to compare the convergence performance of the proposed method with the back/forward sweep method [5,8]. For this purpose, the proposed algorithm is implemented with similar program structures as program DISFLO [8]. Both DISFLO and the proposed method are written in FORTRAN, compiled using SUN FORTRAN 4.1 compiler and run on UNIX system. Similar subroutines

are used to setup the arrays, to order line sections, and to calculate power mismatches. Finally, the same convergence criteria, which is $10^{-6}$, is compared with the maximum P and Q mismatch normalized by the total P and Q flowing into the system from the root node to check convergence. Table 2 shows the number of iterations and solution time for the proposed method and the back/forward method. The elapsed execution time is measured on a SUN Sparc10 workstation. This time includes the execution of all the steps in Figure 3. Table 3 shows the convergence pattern of the proposed method and the back/forward sweep method, where Pm and Qm are the maximum normalized P and Q mismatch respectively.

**Table 2 Comparison of the Proposed Method with Back/forward Sweep Method**

| Test Sys. | No. of Nodes | Proposed Method | | Back/forward Sweep | |
|---|---|---|---|---|---|
| | | No. of Iteration | Time (seconds) | No. of Iteration | Time (seconds) |
| 1 | 490 | 7 | 0.490 | 6 | 0.476 |
| 2 | 722 | 12 | 0.805 | 9 | 0.756 |
| 3 | 1651 | 6 | 1.820 | 6 | 1.799 |

From Table 2 and Table 3, it is seen that the proposed Newton method is as robust as the back/forward sweep method. All the cases tested have reached convergence regardless the wide ranges of r/x ratio and line impedance. For some cases, almost the same number of iterations is needed for both methods. For other cases, slightly higher number of iterations is needed for the proposed method. The following observations are made with respect to this convergence pattern: for distribution systems, slight approximations in the Jacobian matrix seem to have a significant impact on the convergence rate, which, numerically, makes the proposed Newton method performing equally to a non-Newton method, such as the back/forward sweep method in [8]. On the other hand, these small approximations are necessary for the Jacobian matrix to be in the $UDU^T$ form which in turn is necessary for avoiding the explicit formation of the Jacobian matrix, thus avoiding the possible ill-conditions. In this sense, it seems that the proposed Newton method has found the appropriate compromise between numerical efficiency and robustness. With such compromise, it is possible to have a kind of general Newton method for radial power flow solutions.

## V. Other Applications

Although the proposed Newton method is aimed at solving power flow for single-phase (balanced) radial distribution systems without the dispersed generators, it is very general, in the sense that it can be extended to considering many other practical issues. Applications to the following issues are given as examples:

**Table 3 Convergence Pattern of the Proposed Method and the Back/Forward Sweep Method**

| Iteration # | | 490 Nodes System | | 722 Nodes System | | 1651 Nodes System | |
|---|---|---|---|---|---|---|---|
| | | Proposed | B/F Sweep | Proposed | B/F Sweep | Proposed | B/F Sweep |
| 1 | Pm | .29890450 | .29890450 | .14006200 | .14006200 | .43231000 | .43231000 |
| | Qm | .13922300 | .13922300 | .05244800 | .05244800 | .17015400 | .17015400 |
| 2 | Pm | .01788415 | .00643219 | .11854833 | .00982138 | .07051180 | .00981800 |
| | Qm | .01148268 | .01113158 | .08375601 | .02180936 | .04698754 | .02090195 |
| 3 | Pm | .00093628 | .00057982 | .01609456 | .00415140 | .00144529 | .0010447 |
| | Qm | .00087967 | .00027818 | .01513692 | .00148164 | .00063074 | .00036818 |
| 4 | Pm | .00012195 | .00001753 | .00479465 | .00034887 | .00013190 | .00002271 |
| | Qm | .00010286 | .00003554 | .00380562 | .00077424 | .00005399 | .00004796 |
| 5 | Pm | .00001684 | .00000254 | .00134349 | .00015501 | .00001117 | .00000240 |
| | Qm | .00001454 | .00000131 | .00118463 | .00005526 | .00000461 | .00000085 |
| 6 | Pm | .0000247 | .00000009 | .00041353 | .00001314 | .00000096 | .00000005 |
| | Qm | .00000212 | .00000020 | .00035289 | .00002914 | .00000039 | .00000011 |
| 7 | Pm | .00000036 | | .00012524 | .00000585 | | |
| | Qm | .00000031 | | .00010822 | .00000209 | | |
| 8 | Pm | | | .00003833 | .00000050 | | |
| | Qm | | | .00003300 | .00000110 | | |
| 9 | Pm | | | .00001171 | .00000022 | | |
| | Qm | | | .00001010 | .00000008 | | |
| 10 | Pm | | | .00000358 | | | |
| | Qm | | | .00000309 | | | |
| 11 | Pm | | | .00000109 | | | |
| | Qm | | | .00000094 | | | |
| 12 | Pm | | | .00000033 | | | |
| | Qm | | | .00000029 | | | |

- state estimation
- systems with dispersed generators
- systems with loops
- unbalanced systems.

**State Estimation**

The objective of the state estimation is to find the system state $x$ such that the measurement residual $r$ is minimized in a certain sense. In the least square sense, the following system of linear equations is to be solved at each iteration:

$$H^T(x)H(x)\Delta x = H^T(x)r \tag{39}$$

where $H(x)$ is the measurement Jacobian matrix. By using the classical Newton method, the gain matrix $H^T(x)H(x)$ is factorized into $LL^T$ at every iteration, which is time consuming, without mentioning the ill-conditions associated with the Jacobian matrix. By using the proposed method, $H(x)$ can be formed as $MD(x)U^T$, where $M$ is the measurement to branch incidence matrix. Details on the formulation of $M$ can be found in [11]. Now we have:

$$H^T(x)H(x) = UD^T(x)M^T MD(x)U^T \tag{40}$$

Then (39) can be simplified as:

$$M^T MD(x)U^T \Delta x = M^T r \tag{41}$$

and solved in the following steps:

$$M^T My = M^T r \tag{42}$$

$$D(x)U^T \Delta x = y \tag{43}$$

Note $M$ is not a function of $x$ and $M^T M$ needs to be factorized only once. Solving $\Delta x$ from (43) is a forward sweep. Matrix $U$ does not need to be formed.

**Systems with Dispersed Generators**

If the dispersed generator is modeled as a PQ node, little special treatment is needed in the power flow algorithm. However, if the dispersed generator is modeled as a PV node, then for the PV node (say node i), $\Delta Q_i$ is unknown but $\Delta V_i$ is known as zero. Equation (28) has the following form (note that matrix $A$ is not the node to branch incidence matrix here).

$$AX_b = b_x \tag{44}$$

where subscript b in $X_b$ means some of the entries in $X_b$ are known and similarly, subscription x in $b_x$ means some of the entries in $b_x$ are unknown. Note the number of unknown variables in $b_x$ is the same as the number of known variables in $X_b$. Also, these known and unknown pairs are on the same rows of $X_b$ and $b_x$. Of course $AX_b = b_x$ can be rearranged so that all unknown variables are on the left hand side, but this will require factorizing the coefficient matrix, which is expensive. One way to handle this problem is splitting $b_x$ into $b_1$ and $b_2$, and $X_b$ into $X_1$ and $X_2$. $b_1$ includes all the known entries and fills unknown entries with zero. Assume:

$$AX_1 = b_1 \tag{45}$$

$$AX_2 = b_2 \tag{46}$$

then $X_1$ can be solved from (45). In $b_2$, only entries corresponding to the PV nodes are unknowns, all others are zeroes; in $X_2$, only entries corresponding to the PV nodes are known to meet the requirement $\Delta V = 0$. The unknowns in $b_2$ can be solved from the equations with known entries in $X_2$ through (47):

$$Z_v[b_2]_x = [X_2]_b \tag{47}$$

Recalling $A$ has the form $UDU^T$, it can be shown that $Z_v$ is similar to the 'PV node sensitivity matrix' defined in [8] with the line impedances replaced by the equivalent line impedances.

**Systems with Loops**

Assuming node k is the selected breakpoint in a loop and it is broken into nodes i and j, we have the following boundary conditions:

$$\Delta\theta_i - \Delta\theta_j = 0, \qquad \Delta V_i - \Delta V_j = 0$$

$$\Delta P_i + \Delta P_j = \Delta P_k, \qquad \Delta Q_i + \Delta Q_j = \Delta Q_k$$

By performing one row operation on $U$ and one column operation on $U^T$, $UDU^T x = b$ can be transformed into the form of (44) and $x$ in both $x_b$ and $b_x$ can be obtained.

**Unbalanced Systems**

Under the same assumptions as described at the beginning of section III, we can prove that the Jacobian matrix still has the form as $UDU^T$ in a block format. The dimension of each block depends on the number of phases of each node and line.

## VI. Conclusions

This paper presents a modified Newton method for solving the power flow of radial distribution systems. The derivation of this method has revealed that under certain assumptions, the Jacobian matrix of a radial system can be formed as $UDU^T$, with identical topology to that of the Nodal Admittance Matrix. More importantly, such a Jacobian matrix allows a back/forward iteration algorithm to be used instead of

the conventional LU factorization. Thus, the possible ill-conditions associated with the Jacobian and its LU factors are avoided. Tests of the proposed method on large distribution feeders have shown that it is as robust and efficient as the back/forward sweep method.

Since the proposed method is based on the Newton formulation, the experiences people have accumulated over the last 30 years of using the Newton method in transmission systems can be shared in distribution systems. An outstanding example is in the area of distribution state estimation.

Although the proposed Newton method is aimed at solving power flow for single-phase (balanced) radial distribution systems without dispersed generators, it can be extended to the solution of systems with loops, dispersed generators, and three phase (unbalanced) representation.

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## Appendix A Basic Electric Circuit Theory Review

For a linear, time-invariant RLC circuit with sinusoidal voltage, which a typical distribution system is modeled as, the ohm's law has the following familiar form:

$$\dot{I}_b = Y_b \dot{V}_b \tag{A.1}$$

where $\dot{I}_b$ : a vector of complex branch current,

$Y_b$ : a diagonal matrix with diagonal element to be branch admittance,

$\dot{V}_b$ : a vector of complex branch voltage.

The directions of $\dot{I}_b$ and $\dot{V}_b$ must be consistent. For KCL, we have:

$$A\dot{I}_b = \dot{I}_n \tag{A.2}$$

where $\dot{I}_n$ : vector of nodal injection current,

$A$ : node to branch incidence matrix, defined as:

$$A_{ij} = \begin{cases} 1, \text{ if branch j is directed away from node i} \\ -1, \text{ if branch j is directed towards node i} \\ 0, \text{ if branch j is not incident to node i} \end{cases}$$

Each row of (A.2) corresponds to KCL for a node. Since each branch is always directed away from one node and towards one other node, the summation of all rows will end with zero. Hence, for a system with n nodes, only n-1 rows in (A.2) are independent. An arbitrary row can be removed from (A.2) so that the remaining n-1 rows are independent. The node corresponding to the removed row is usually called the reference node.

To express KVL in matrix form, a loop matrix $B$ is defined as:

$$B_{ij} = \begin{cases} 1, \text{ if branch j is in loop i, and their dir. agree} \\ -1, \text{ if branch j is in loop i, and their dir. oppose} \\ 0, \text{ if branch j is not in loop i} \end{cases}$$

then KVL can be expressed as:

$$\boldsymbol{B}\dot{\boldsymbol{V}}_b = \boldsymbol{0} \qquad \text{(A.3)}$$

Each row of (A.3) corresponds to KVL for a loop. It is not necessary to include all loops in (A.3) since independent loops are sufficient for $\dot{V}_b$ to obey. The independent loop is defined as that not all the branches of the independent loop can be found in other independent loops.

## Biographies

**Fan Zhang** (M'94) was born in Fuzhou, China. He received his B.S. from Zhejiang University, China, in 1983, M.E. from Manhattan College, Bronx, NY, in 1989, and Ph.D. from Georgia Institute of Technology in 1994. All the degrees are in Electrical Engineering. From 1983 to 1987, he was employed as an analysis engineer by Fujian Power Control Center, China. In April 1994, he joined the ABB Automated Distribution Division as a Power Systems Engineer responsible for developing and enhancing two software packages CADPAD® (Computer-Aided Distribution Planning And Design) and CADOPS™ (Computer-Aided Distribution Operation Planning System).

**Carol S. Cheng** (M'92) was born in Beijing, China. She received her B.S. in Electrical Engineering from Northern Jiaotong University in Beijing in 1982, M.S. in Mechanical Engineering from the University of Cincinnati in 1986, and Ph.D. in Electrical Engineering from Georgia Institute of Technology in 1991. From 1982 to 1985, she worked for Northern Jiaotong University as a Teaching Assistant. In 1992 she joined PG&E as a Consulting Engineer, responsible for developing real-time applications for distribution systems. In 1995 she became a Systems Engineer/Operational Analyst, responsible for developing tools and applications to support power system planning and operation.

DISCUSSION

ELHAJ BARAKAT and S.A. AL-RASHED (Saudi Consolidated Electric Company, Riyadh, Saudi Arabia): We wish to commend the authors for their valuable contribution in clearing the misconception that the Newton method is not appropriate for radial distribution power flow analysis. The test results on the convergence performance of the proposed method in comparison with recently published back/forward sweep methods have shown that the proposed method is equally robust and efficient. However, we would like to have authors opinion and clarification on the following points:

1. Although the proposed method is primarily aimed at solving power flow in single-phase radial distribution systems it would have been useful if the authors could present quantitative analysis showing sample comparative results with reference to most common practical distribution systems and in particular unbalanced 3-phase systems with loops.

2. Although the authors have selected the convergence criterion as a means of comparing the performance of the proposed method relative to that of the back/forward sweep methods, it would be useful to show, if any, cases where the proposed method performance is better or worse than the later methods. This would help future researchers in this area.

3. In Section IV, the authors stated that for some other cases, slightly higher number of iterations is needed for the proposed method. It would be useful if the authors throw some light on these cases.

Finally, we would look forward to authors' further investigation in load flow analysis in more practical distribution systems.

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**Fan Zhang and Carol S. Cheng**: We would like to take this opportunity to thank the discussers for their kind remarks and for their interests in our paper. First, we would like to clarify that all test systems we used are practical distribution systems. The points raised by the discussers are answered in the following:

Even though the single phase radial distribution systems are used to express the concepts of the modified Newton method, to apply the same concept to the unbalanced 3-phase systems with loops is also possible as shown in the paper. We are sorry we don't have any quantitative results to show at this point. But we encourage people to investigate this and compare the results with that obtained from the back/forward sweep method.

Based on our test results, the convergence rate of the proposed method is always slightly slower than the back/forward sweep method. We believe that the strength of the proposed method is not purely its speed, but its ability to use the Jacobian matrix and the back/forward sweep algorithm at the same time, so that a more general Newton's formulation can enjoy almost the same convergence rate as the back/forward sweep method.

The variation of the number of iterations for the different test cases is due to the approximation of the Jacobian matrix. The difference between the Jacobian matrix and its approximation is the function of system parameters including system topology. For one system, this difference might be greater than the other systems, hence the slower convergence rate.

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